

LS-Face: Hybrid Cascade Model Design Document

Authors: Computer Vision Research Group 3 (USLS) **Project:** Smart Gate Face Recognition System **Target Platform:** Edge Hardware (Raspberry Pi 5 with Optional Accelerator)

The **LS-Face Hybrid Cascade Model** routes every frame through a fast classical recognizer (**LBPH**) first, and escalates only the frames that need it to a deep recognizer (**SFace**). This document specifies the pipeline stages, the escalation gate that decides when to escalate, the tunable thresholds, and the target hardware layout.

1. System Architecture and Pipeline Stages

The pipeline has three stages: **shared detection/alignment (YuNet)**, **Stage 1 (classical fast path: LBPH)**, and **Stage 2 (deep escalation: SFace)**.

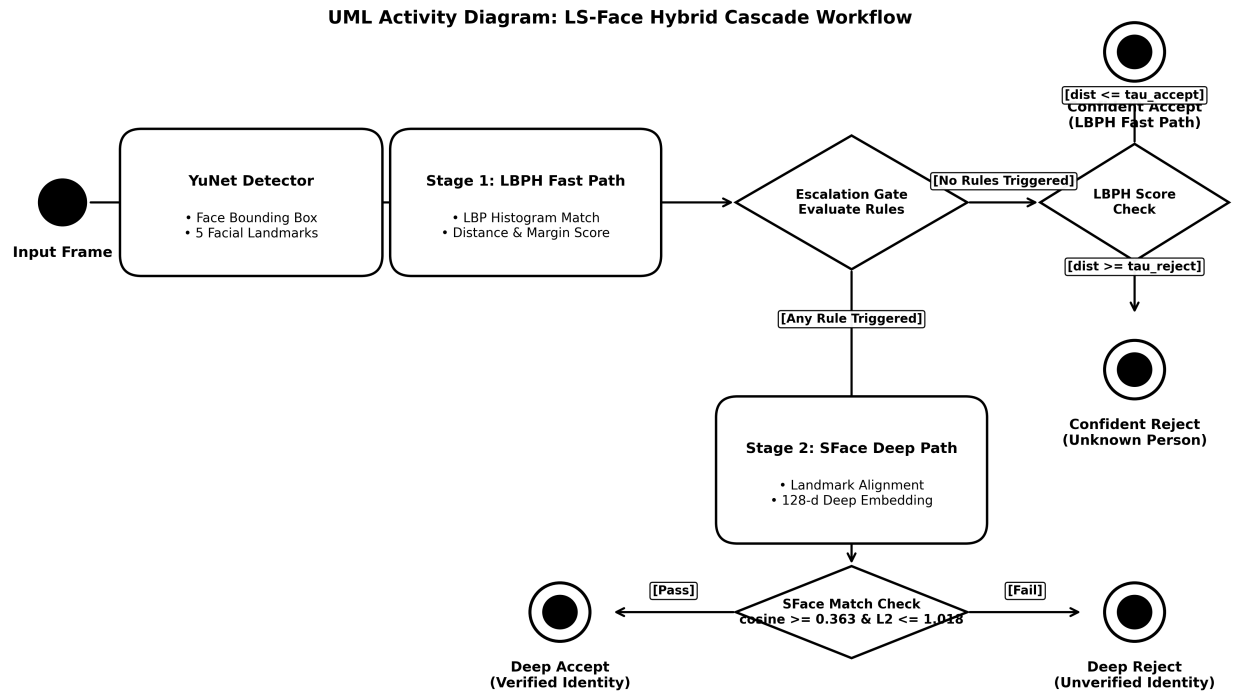


Figure: LS-Face Hybrid Cascade Model Workflow Diagram

1.1 Front-End Detector (YuNet)

Before face recognition happens, a lightweight neural detector called **YuNet** inspects the raw camera frame.

- **Face Detection:** Locates faces and extracts bounding box coordinates.
- **Facial Keypoint Landmark Alignment:** Identifies 5 key facial points (left eye, right eye, nose tip, left mouth corner, right mouth corner).

- **Shared Efficiency:** YuNet outputs are shared by both recognizers, avoiding duplicate face detection operations.

1.2 Stage 1: Classical Fast Path (LBPH Recognizer)

The detected face crop is sent directly to the **LBPH** (Local Binary Patterns Histograms) recognizer on the CPU.

- **Feature Extraction:** Analyzes local texture patterns across small regions of the face.
- **Nearest Neighbor Matching:** Calculates the distance score to enrolled identities. A smaller distance score indicates higher similarity.
- **Relative Margin Check:** Compares the distance score of the top match against the second closest match to confirm clear separation.

1.3 Stage 2: Deep Feature Escalation (SFace Recognizer)

When the Escalation Gate determines that a face probe needs deeper inspection, execution moves to **SFace**.

- **Landmark-Based Alignment:** Uses the 5 facial keypoints from YuNet to warp and orient the face crop into a clean, standardized format.
- **Deep Feature Extraction:** Converts the aligned face into a 128-element feature vector representing unique facial characteristics.
- **Similarity Matching:** Measures similarity using Cosine similarity and Euclidean distance against enrolled gallery feature vectors.

2. The Escalation Gate and Quality Probes

The **Escalation Gate** acts as the intelligent controller of the system. It determines whether LBPH's decision is confident enough to accept, or if SFace must take over.

UML Decision Diagram: Escalation Gate Evaluation Logic

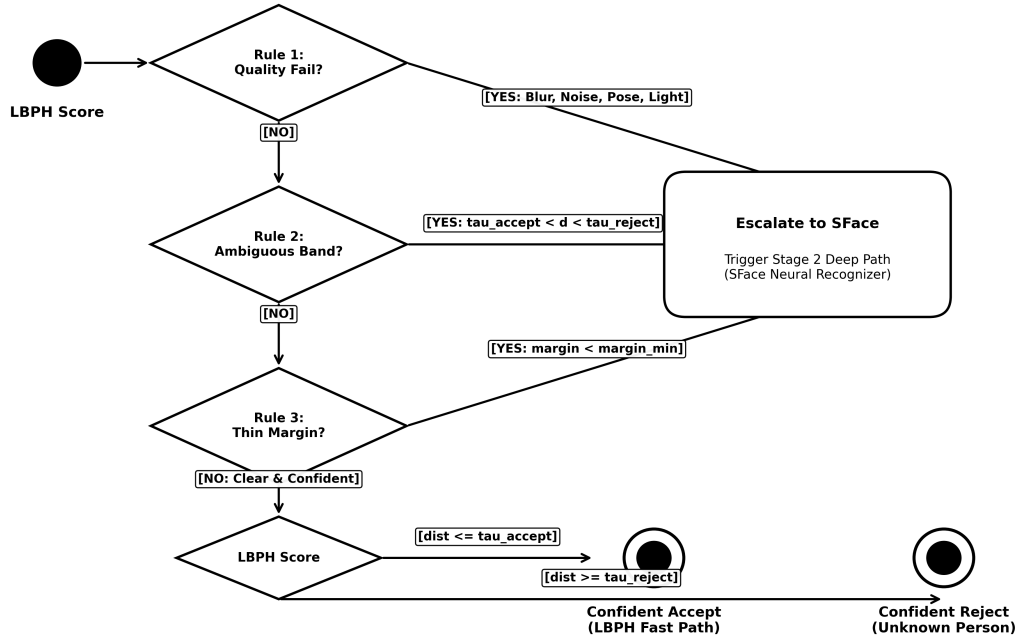


Figure: Escalation Gate Decision Logic Flowchart

2.1 Escalation Rules

The gate evaluates three rules in order, and escalates to SFace on the first match:

#	Rule	Trigger	Rationale
1	Quality Flag Override	Any quality probe fails (blur, light, noise, pose, size)	Quality checks take priority, as LBPH is unreliable in poor conditions
2	Ambiguous Distance Band	$\text{tau_accept} < \text{distance} < \text{tau_reject}$	LBPH's own score is inconclusive
3	Thin Separation Margin	$\text{margin} < \text{margin_min}$	Top-1 and top-2 matches are a near-tie

If none of the rules trigger, the gate resolves the frame on the fast path alone:

- $\text{distance} \leq \text{tau_accept} \rightarrow$ **Confident Accept**
- $\text{distance} \geq \text{tau_reject} \rightarrow$ **Confident Reject** (unknown person)

2.2 Real-Time Quality Probes

Five lightweight quality probes run during the fast path to check image clarity:

- **Motion Blur Probe:** Measures image sharpness using variance of Laplacian. Images below the blur threshold are marked as blurry.
- **Illumination Probe:** Measures average image brightness (luma). Images below the low-light threshold or above the bright-light threshold are marked as poorly lit.

- **Noise Probe:** Estimates high-frequency sensor noise. Images above the noise threshold are marked as noisy.
- **Off-Pose Angle Probe:** Uses facial landmark locations to check head tilt and turn angle. Faces turned beyond the pose threshold are marked as off-pose.
- **Face Size Probe:** Checks the pixel height and width of the face bounding box. Faces smaller than the size threshold are marked as small faces.

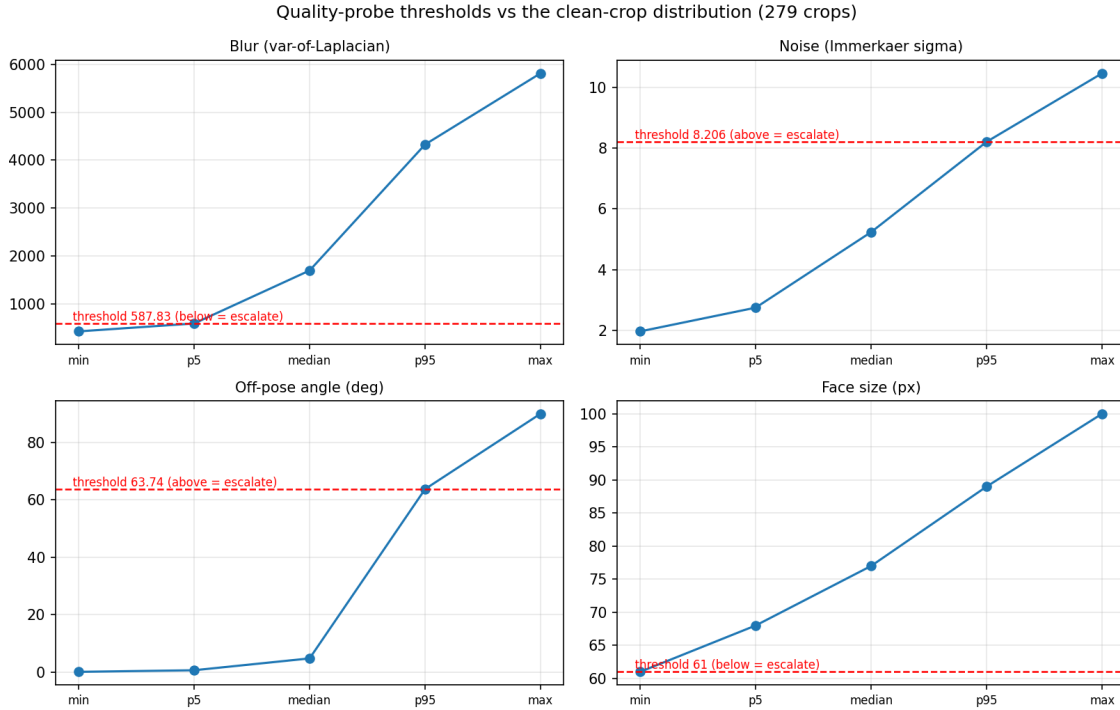


Figure: Quality Probe Distributions and Threshold Edges

3. System Thresholds and Configurations

The hybrid model uses calibrated thresholds to control gate routing and matching decisions.

Parameter	Function	Scope and Purpose
tau_accept	LBPH Accept Distance	Distance score threshold for confident fast path acceptance
tau_reject	LBPH Reject Distance	Distance score threshold for non-enrolled unknown rejection
margin_min	Top-1 vs Top-2 Separation	Minimum relative gap required between best and second-best match
sface.l2_genuine	SFace Matching Distance	Maximum Euclidean distance for genuine deep feature verification
sface.cosine_genuine	SFace Similarity Floor	Minimum Cosine similarity score for genuine deep feature verification
tau_blur	Blur Threshold	Minimum required sharpness value before triggering blur flag

Parameter	Function	Scope and Purpose
luma_lo / luma_hi	Light Range	Acceptable brightness range before triggering illumination flag
tau_noise	Noise Threshold	Maximum tolerable noise level before triggering noise flag
tau_pose	Pose Angle Limit	Maximum head tilt/turn angle before triggering off-pose flag
px_min	Minimum Face Size	Minimum bounding box size required for classical processing

4. Edge Hardware and Software Architecture

4.1 Deployment on Edge Platforms

The system is tailored for single-board computers like the Raspberry Pi 5.

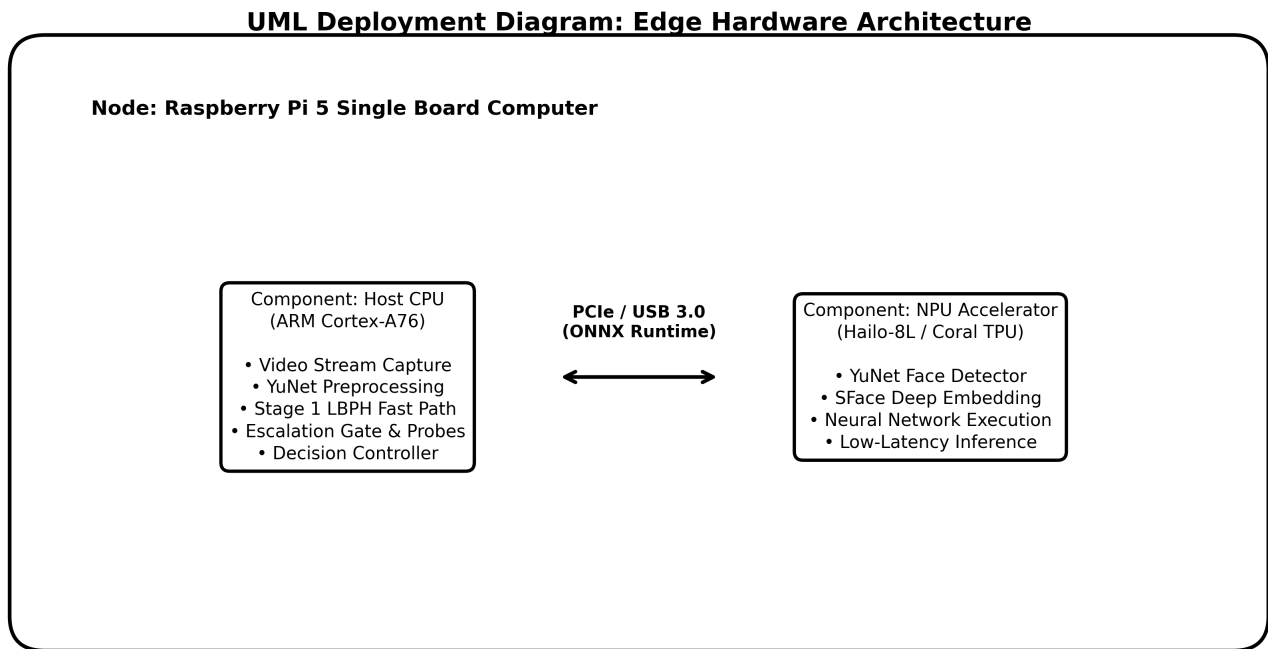


Figure: Edge Hardware Deployment Architecture

- **Host CPU (ARM Cortex-A76):** Manages video capture, YuNet preprocessing, LBPH fast path execution, real-time quality probing, and gate control logic.
- **Accelerator / NPU (Optional):** Runs YuNet face detection and SFace deep feature extraction efficiently when available.

4.2 Operating Modes and Fallback Protection

The system supports three software operating modes:

- **Cascade Mode (Default):** Full hybrid operation with dynamic quality checking and score gating.

- **CV-Only Mode (Fallback):** Runs purely on LBPH. If deep learning models or neural runtime libraries are missing, the system falls back gracefully to CPU-only operation without crashing.
- **DL-Only Mode (Comparison):** Bypasses LBPH and runs deep feature extraction on every frame for baseline testing.